

OSSP (Operating Systems And Systems Programming) - 25CS2104E

SKILL -1

QUESTION

Install Linux VM, configure GCC, set up a Git repository and project structure, understand shell architecture, build an initial Makefile, and analyze fork()/exec() and parent-child processes.

Simple C Program

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child PID: %d\n", getpid());
        execlp("ls", "ls", NULL);
    } else {
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
    }
    return 0;
}
```

Output:

```
pothubhavana@DESKTOP-16SJJUA:~$ nano skill1.c
pothubhavana@DESKTOP-16SJJUA:~$ gcc skill1.c
pothubhavana@DESKTOP-16SJJUA:~$ ./a.out
Parent PID: 572
Child PID: 573
a.out  addition.c  exc.c      files.c    open.c    p3.c      process.c  skill1.c   systems  week2.c
add    create.c     excess.c   files.c.save  p2       p4.c      process.c.save  sub        t2.txt   week3.c
add.c  diri        exworking.c fork.c     p2.c     parent.c  read.c.save  subtraction.c week1.c   zombie.c
pothubhavana@DESKTOP-16SJJUA:~$
```

